

Abstract

Medical-grade electroencephalography (EEG) systems are impractical for usage in self-directed research, such as sleep stage classification; however, the data reliability and scope for lower-cost alternatives (e.g. OpenBCI Ganglion) require further validation. To observe motor artifacts and alpha blocking (the decrease in alpha waves during the transition from eyes closed to eyes open^[1]), controlled experiments were conducted with and without conductive paste. Additionally, a Python-based pipeline was developed to classify sleep stages (Awake, N1, N2, N3). This exploratory study investigated the role of electrode paste, signal reliability, sleep stage classification, scope of generative AI for as a tool for self-directed research.

Background & Motivation

- Consumer-grade devices use fewer channels and use dry electrodes that are susceptible to a low signal to noise ratio (SNR), limiting ability to provide trustworthy data due physiological noise (EOG/EMG artifacts).^[4]
- Traditional sleep stage classification often does not fit into rigid scoring rules and researchers are moving towards data-driven approaches.^[1]
- Each sleep stage has distinct EEG biomarkers. Wakefulness is marked by a transition from beta to alpha waves, theta waves for N1, spindles and K-complexes for N2, and delta waves for N3.^[3]

Methods

EEG signals were collected using a four-channel OpenBCI Ganglion operating at a sampling rate of 200 Hz.^[2] All data was self-collected from the researcher (Disha) across 40 trials for data reliability and four, 30-70 minute sleep sessions.

Based on the international 10-20 system, flat electrodes were placed at the frontal (Fp1, Fp2) region and comb electrodes were placed at the occipital (O1, O2) region, with references nodes at A1 and A2. An alcohol wipe was used to prep the skin surface prior to electrode placement. The electrodes were secured to the head using the OpenBCI headband kit to maintain sensor contact during trials. All trials were conducted in a dark, quiet, controlled environment.

To examine the reliability of the EEG signals and the role of the electrode paste, four simple experiments with 10 trials each were conducted: eyes open v. eyes closed and tapping v. resting, each with and without electrode paste. Alpha/motor ratios were calculated for each trial and averaged.

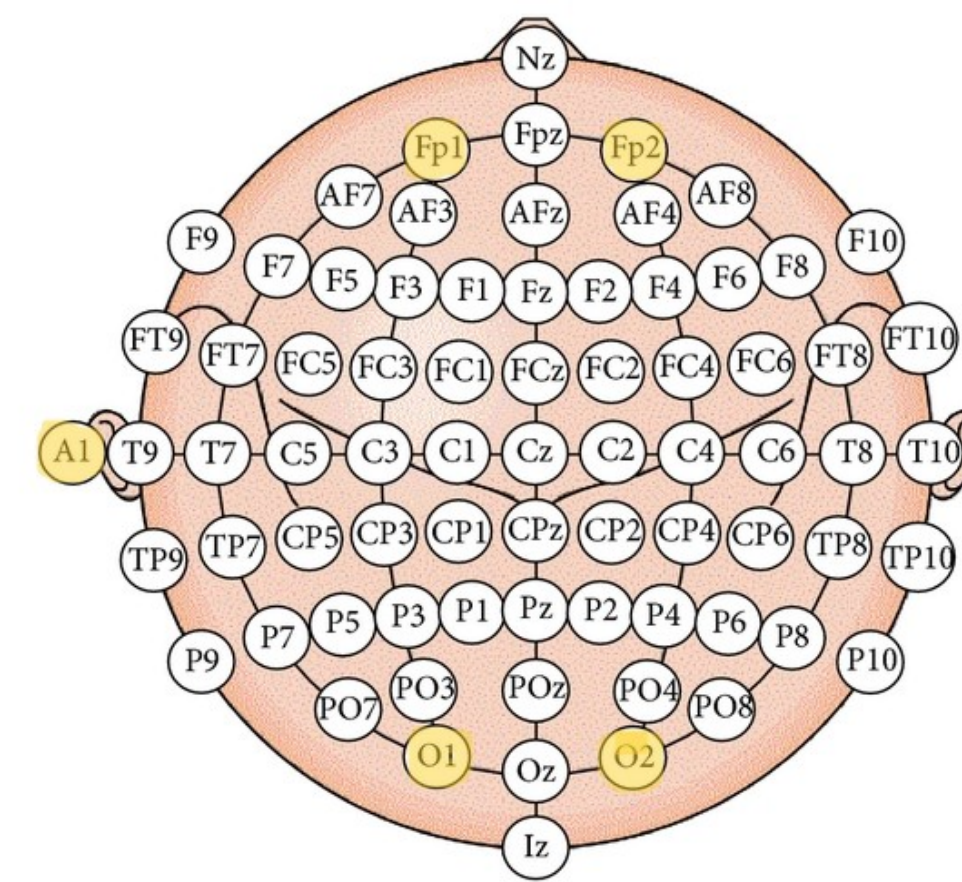


Figure 1. Electrodes (10-20 system) measured and secured at the highlighted locations (Fp1, Fp2, O1, O2), with A1 and A2 attached to earlobes.^[2]

Sleep Stage Classification

The Python-based pipeline was developed using ChatGPT and Gemini Pro to investigate the role of AI in developing more robust signal processing algorithm to aid preprocessing and classification for student research initiatives. Raw CSV files recorded with OpenBCI GUI were analyzed.

- 4th-order Butterworth band-pass filter (0.5 – 45 Hz) to eliminate slow eye movements and triboelectric noise.
- Artifacts were recorded if peak-to-peak amplitude < 400 μ V.
- Rule-based sleep stage classification algorithm developed based on American Academy of Sleep Medicine (AASM) standards.^[1]
- Dynamic transition hierarchy to account for natural progression through sleep stages^[3]
- Rule-based smoothing using a 7-epoch window to minimize unrealistic transitions

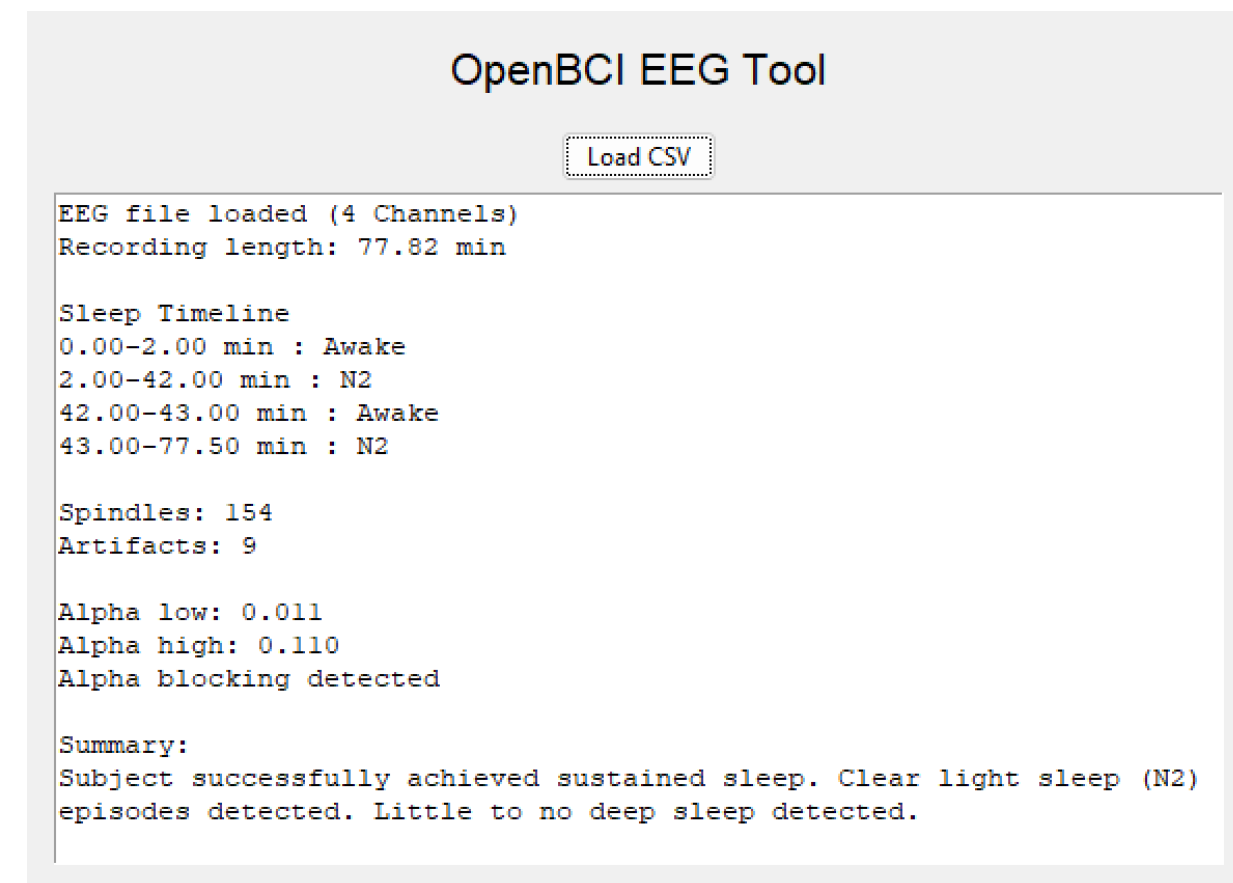


Figure 2. A summary of the key features of the sleep trial generated by the Python script. Includes the length of the trial, the different stages detected, spindle & artifact counts, alpha blocking, and a text summary.

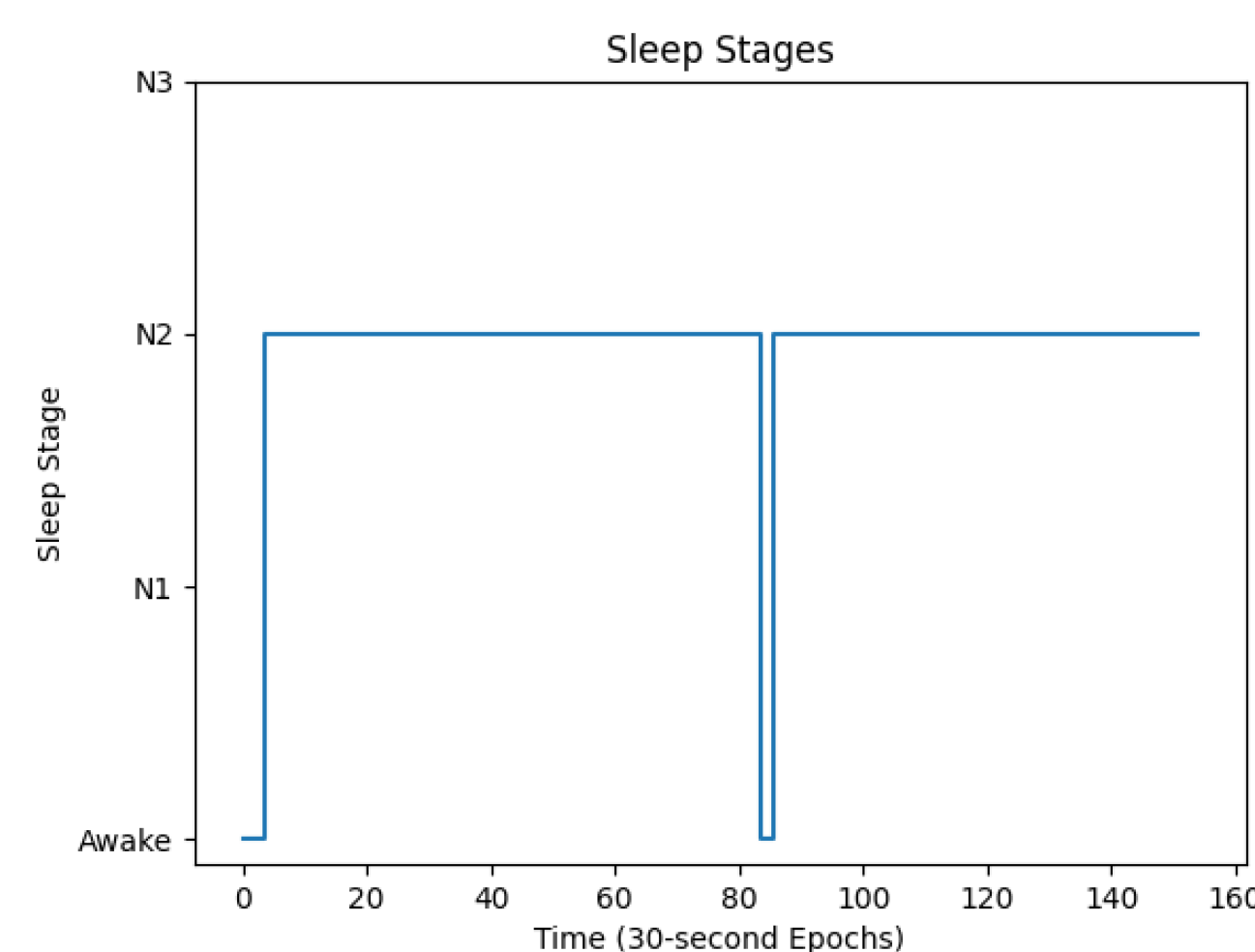


Figure 4. A simplified hypnogram tracking sleep stages for the trial. N1 is a brief period of light sleep, N2 is deeper sleep characterized by sleep spindles, and N3 is deep sleep or slow-wave sleep.^[3] The recording demonstrates mostly N2 sleep with brief windows.

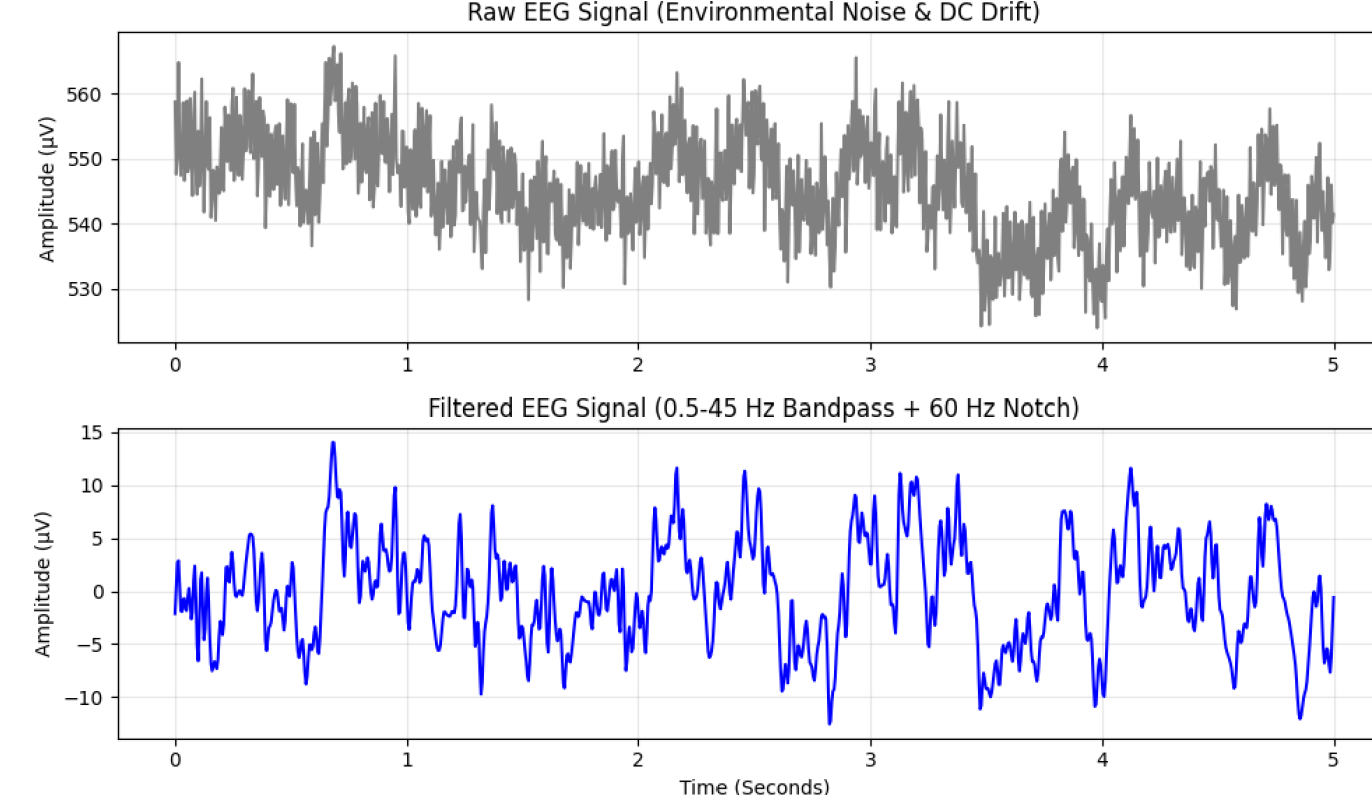


Figure 3. A comparison of the raw and filtered EEG data. The filtered signal is within a realistic -15uV to 15uv range with sharper waveforms.

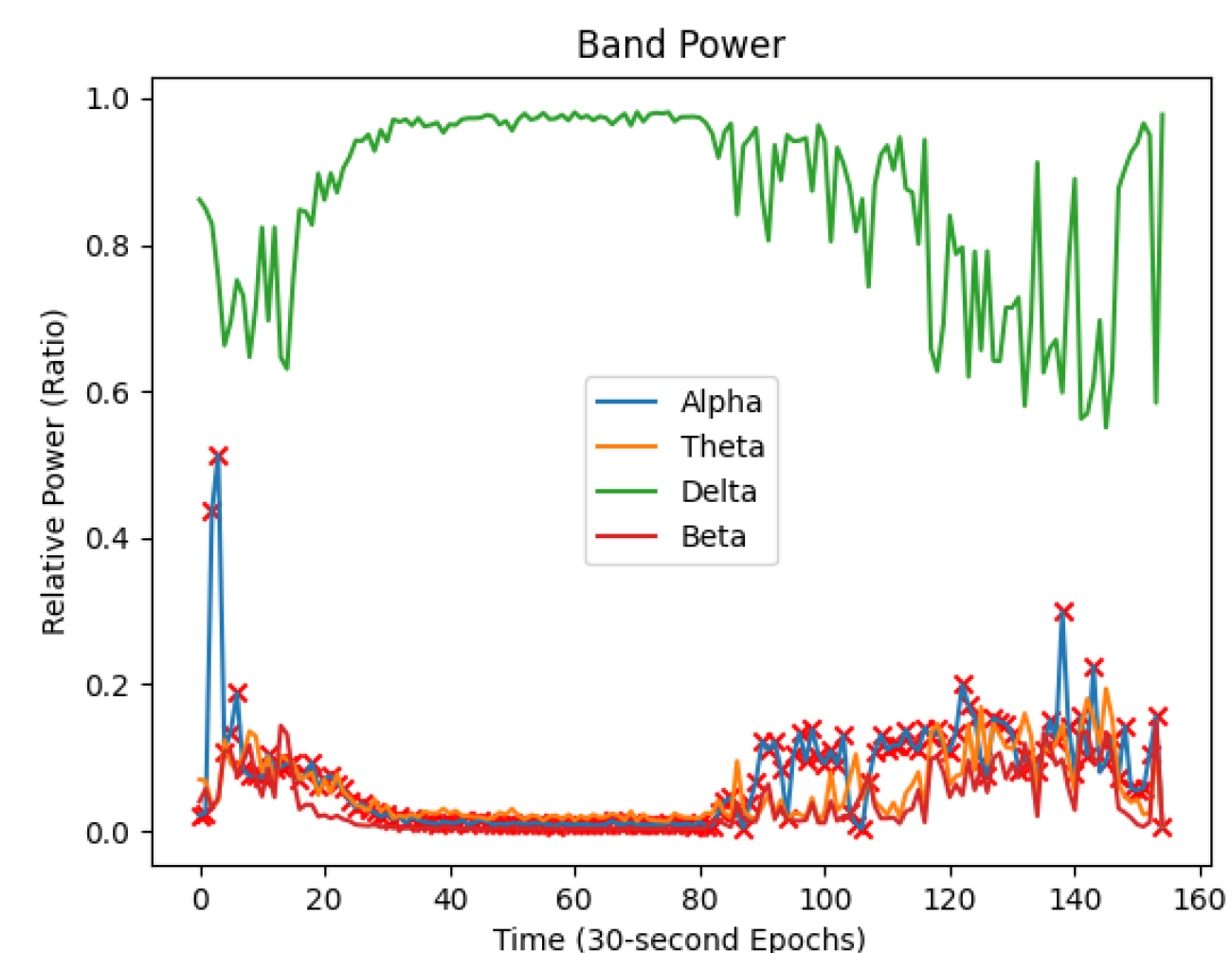


Figure 5. The relative band power ratios for alpha, theta, delta, and beta waves. The 'X' is an indicator of a spindle. Delta waves are dominant during N2 phase, with peaks in alpha demonstrating eyes closing after brief awakenings.

Data Reliability Analysis

Averages for Eyes Closed v. Eyes Open	
	Alpha Ratio (Closed/Open)
No-Paste	0.97
Paste	8.88

Table 1. A comparison of the alpha band power during eyes open and eyes closed states. Alpha ratio must be >1 to indicate a decrease in alpha when transitioning from eyes closed to open.

Averages for Tapping v. Resting		
	Motor Ratio (Rest/Tap)	
	Alpha	Beta
No-Paste	1	1.08
Paste	2.45	0.87

Table 2. A comparison of alpha and beta band power during tapping and resting states. The paste trials successfully captured standard motor behavior, showing the expected decrease in alpha waves (ratio >1) and increase in beta waves (ratio <1) during tapping, which is not reflected for the no-paste experiment.

Conclusion

- EEG signal quality and reliability is improved with the usage of electrode paste.
- Sleep stage classification achieved, particularly spindle count, awake, and N2 sleep stages. Further investigation is required to determine if N3 can be distinguished; the current algorithm fails to detect the N1 phase.
- Generative AI tools such as ChatGPT and Gemini can be used in self-directed research to develop algorithms and seek clarification on research.

Future Direction

- Extended Monitoring: Conduct multi-hour sleep trials to better capture and investigate the detection of N3 sleep.
- Algorithm: Develop machine learning frameworks to develop an advanced AI-driven algorithm, improving the classification accuracy of transitional N1 sleep.
- Participants: Scale the participant pool to increase statistical power and achieve higher confidence intervals across all findings.

References

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